Zekun Qi

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Current research directions include 3D computer vision, multimodal large language models, and embodied AI.

Education

2022 – · · · ·
M.S. Xi'an Jiaotong University & IIIS, Tsinghua University *Software Engineering*
Supervised by Prof. **Andrew C. Yao** and Prof. **Kaisheng Ma**, while also collaborating closely with Prof. **Li Yi** and Prof. **He Wang**. Our group's homepage is **ArchipLab**.

2018 – 2022
B.S. Xi'an Jiaotong University *Automation*
Ranked in the top **10%** academic performance. Achieved a **first-class** university scholarship and honored as an outstanding graduate.

Employment History

2024.3 – · · · · · ·
Galbot Inc. *Research Intern*
Embodied LLM structure design for open vocabulary robotic manipulation.

2023.7 – 2024.3
MEGVII Inc. *Research Intern*
Multi-modal Large Language Model structure design and data filtering.

2021.6 – 2022.2
ByteDance Inc. *Engineer Intern*
Build a performance monitoring platform using Selenium. Use SIFT and KNN algorithms to achieve online front-end comparison. Construct the Lark service bot.

Research Publications

Conference Proceedings

1. **Z. Qi**, R. Dong, G. Fan, Z. Ge, *et al.*, **Contrast with reconstruct: Contrastive 3d representation learning guided by generative pretraining**, in **ICML**, **2023**. URL: https://openreview.net/forum?id=80IfYewOh1.
2. **Z. Qi**, M. Yu, R. Dong, and K. Ma, **Vpp: Efficient universal 3d generation via voxel-point progressive representation**, in **NeurIPS**, **2023**. URL: https://openreview.net/forum?id=etd0ebzGOG.
3. **Z. Qi**, R. Dong, S. Zhang, H. Geng, *et al.*, **Shapellm: Universal 3d object understanding for embodied interaction**, in **European Conference on Computer Vision**, **2024**. URL: https://arxiv.org/abs/2402.17766.
4. R. Dong, **Z. Qi**, L. Zhang, J. Zhang, *et al.*, **Autoencoders as cross-modal teachers: Can pretrained 2d image transformers help 3d representation learning?** In **ICLR**, **2023**. URL: https://openreview.net/forum?id=8Oun8ZUVe8N.
5. R. Dong, C. Han, Y. Peng, **Z. Qi**, *et al.*, **Dreamllm: Synergistic multimodal comprehension and creation**, in **ICLR Spotlight**, **2024**. URL: https://openreview.net/forum?id=y01KGvd9Bw.

Preprint Articles

1. G. Fan, **Z. Qi**, W. Shi, and K. Ma, **Point-gcc: Universal self-supervised 3d scene pre-training via geometry-color contrast**, *arXiv preprint*, 2023. URL: https://arxiv.org/abs/2305.19623.
2. Y. Peng, Y. Cui, H. Tang, **Z. Qi**, *et al.*, **Dreambench++: A human-aligned benchmark for personalized image generation**, *arXiv preprint*, 2024. URL: https://arxiv.org/abs/2406.16855.

Skills

Languages	reading, writing and speaking competencies in English and Chinese.
Coding	**Python**, C, C++, Matlab, LaTeX, …
Library	**PyTorch**, Numpy, Pandas, OpenCV, sklearn…
Misc.	**ChatGPT**, Linux, **DataBase** (SQL, MongoDB), **Data Mining** (Beautiful Soup, Selenium), Embedded system (Keil-ARM, C51), Android Development, PhotoShop, Blender, …

Miscellaneous Experience

Awards and Achievements

2020

- **National Special Prize for China Undergraduate Physics Tournament**
  Won the **national runner-up** of the China Undergraduate Physics Tournament (CUPT) as the team leader. **Commendation Website**. The **paper** is published in the core journal *Physics and Engineering*.
- **Provincial First Prize for China Undergraduate Mathematical Contest in Modeling**
  Based on graph theory and reinforcement learning to model and analyze desert gold-digging games, the **paper** is published in the core journal *China Collective Economy*.

2021

- **National Excellence Award for Undergraduate Innovation & Entrepreneurship Projects**
  Regional intrusion and abnormal behavior detection, utilizing frame differencing for motion region detection and segmentation, mean shift algorithm for object tracking, and YOLO for object recognition.
- **Annual Spiritual Civilization Award in Xi'an Jiaotong University**
  Won the Innovation and Entrepreneurship Award, which is granted to only **ten** individuals each year. **Commendation Website**.